Giving Instructions Names · AS · Practice

U05d · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

Fresh authored questions. Use your board's pseudocode. No Python or computer is needed. Assume all inputs match their declared types.

1. Ours: choose the type

Give the most suitable type for each: (i) number of parcels; (ii) mass 2.75 kg; (iii) whether the door is locked; (iv) a valid delivery date. **[4]**

2. Ours: declare, then initialise

Declare DeliveryDate as DATE and assign 18/05/2027. Use dd/mm/yyyy. **[3]**

3. Ours: what can change?

Name the programming feature used for a value fixed during execution. **[1]**

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4. Ours: read forward

```
DECLARE Count : INTEGER
DECLARE Total : INTEGER
Count ← 4
Total ← Count * 6
Count ← Count + 3
OUTPUT Total
```

(a) State Count and Total after the fourth line. **[2]**

(b) State Count after the fifth line. **[1]**

(c) State the output of the sixth line. **[1]**

5. Ours: the information must come in

```
DECLARE Parcels : INTEGER
OUTPUT "How many parcels?"
OUTPUT Parcels
```

(a) Write the one statement to insert between the two OUTPUT statements so the entered number is stored. **[1]**

(b) Explain why displaying the question alone does not give Parcels a value. **[1]**

(c) With your repair and input 5, state the two lines displayed, in order. **[2]**

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6. Ours: one fixed price

(a) Write a constant declaration for UnitPrice with the literal value 6. Use your board's syntax. **[1]**

(b) Give two maintenance benefits of using UnitPrice instead of repeating 6 in parcel-cost calculations. **[2]**

(c) Is `CONSTANT DoublePrice = UnitPrice * 2` a valid constant declaration under your guide? Explain. **[1]**

7. Ours: one letter can still be a string

(a) State the types of `'P'` and `"P"` in this notation. **[2]**

(b) Which of CHAR and STRING can hold an empty value? Explain why. **[2]**

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8. Ours: the parcel desk · pen first

A desk charges 6 currency units per parcel. Read a customer label and a valid non-negative whole-number parcel count. Calculate and output the label and total. Use a named constant for the price and declare all variables. No choices or loops are needed. **All six marks are human marked.**

(a) Draw or number the steps before writing statements. **[2]**

(b) Write the pseudocode sequence, including declarations. **[3]**

(c) For customer label Visitor and count 5, state the output. **[1]**

Above and beyond · outside the 30 marks. A and B are INTEGER variables with values 4 and 9. Using a third INTEGER variable, swap their values without losing either. Declare the extra name and trace every assignment. Why does A ← B followed by B ← A fail?

Additional space - label the question number.